

Long-Context Extension Methods for Pretrained Language Models

From Positional Interpolation to KV-Cache and Systems

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Abstract

Context windows advertised on pretrained language models grew from 4–8k tokens in 2022 to a claimed 1–10M by 2026, yet the techniques behind that growth are routinely conflated and their benchmark numbers reported without a common frame. Existing surveys catalogue methods by family—positional interpolation, sparse attention, KV-cache compression, retrieval-in-context, recurrent hybrids—but stop short of asking which family now carries the field’s measurable gains. We organise the entire literature around three root constraints any long-context method must attack: positional extrapolation (R1), attention compute $O(n^2)$ (R2), and KV-cache memory $O(n)$ (R3), and derive the taxonomy from that decomposition rather than asserting it. Surveying 169 papers, tech reports, and model cards across 2022–2026, we compare the sixteen most influential methods along nine dimensions, separating each method’s *claimed* window from its *effective* context length. Our headline finding is that long-context progress is no longer bottlenecked by positional-encoding interpolation; effective context length remains capped by attention allocation and long-data scarcity rather than by position encodings, so measurable gains have shifted to KV-cache management and inference-systems engineering. The persistent gap between claimed and effective windows, its insensitivity to the positional recipe used, and the convergence of frontier 1M-token systems onto stacked KV and serving techniques rather than novel positional tricks all support this reading and reframe the field’s open problems around attention, evaluation, and data.

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1 Introduction

In 2022, a context window of 2k–8k tokens was the norm, and a single long document or codebase simply did not fit; by 2026 the advertised ceiling had climbed three orders of magnitude (Figure 1). This expansion was not one breakthrough but an arms race among several method families that competed and then, increasingly, converged into stacked recipes. The purpose of this survey is not to catalogue them in isolation but to ask a sharper question: where do the real wins now come from? We answer it by forcing every method onto a single comparison frame and separating what a method *claims* from what it *demonstrably* delivers.

1.1 Why Now?

Three concurrent shifts make this the right moment to take stock. First, positional extrapolation became cheap: rotary-embedding interpolation now extends context, often with little or no training. Second, attention and memory became the binding economic problems: dense attention costs $O(n^2)$

at prefill and the key–value cache grows $O(n)$ at decode, so serving a long window—not opening one—is the expense that scales. Third, length-controlled evaluation matured, making it possible to ask not merely whether a model accepts a long input but whether it uses one. We organise the whole survey around the three root constraints these shifts expose: **(R1)** positional extrapolation, **(R2)** attention compute $O(n^2)$, and **(R3)** KV-cache memory $O(n)$, and we derive the taxonomy of six families (Figure 2) from that decomposition rather than asserting it. The same lens reads the field’s trajectory (Figure 3): the dominant lever migrates from positional encoding toward KV-cache and inference systems as windows grow.

1.2 Relationship to Existing Surveys

Several surveys cover adjacent ground; our delta is the root-cause frame and the claimed-versus-effective axis. Zhao et al. [2024] survey length extrapolation from the positional-encoding perspective; we subsume that lineage as our R1 family and argue it is no longer the binding constraint, a claim outside their scope. Huang et al. [2023] review long-context Transformer architecture broadly; we narrow to pretrained decoder-only models and add a normalised claimed-versus-effective comparison. Li et al. [2024a] catalogue KV-cache management; we embed that catalogue inside a thesis about *why* KV work, not positional work, carries the gains. Elouargui et al. [2023] survey efficient Transformers for the R2 compute axis, and Zeng et al. [2024] frame long-context serving as a CAP-style trade-off; we connect both to the production recipes of Section 5. None separates advertised from effective context as the organising measurement.

1.3 Thesis and Contributions

Our thesis is that long-context progress is no longer bottlenecked by positional-encoding interpolation; effective context length remains capped by attention allocation and long-data scarcity rather than by position encodings, so measurable gains have shifted to KV-cache management and inference-systems engineering. We steelman the opposite—that positional interpolation plus continued pretraining is still the highest-leverage path—before rejecting it. This survey contributes:

1. **Root-Cause Frame.** We reduce long-context extension to three constraints (R1, R2, R3) and derive a six-family taxonomy from them, stating for each method what it attacks and what it trades away (§2).
2. **Claimed-vs-Effective Analysis.** We separate advertised windows from effective context length across the corpus and show the gap does not track the positional recipe (§6).
3. **Normalised Comparison Matrix.** We compare sixteen influential methods along nine dimensions in a single cross-cutting matrix (§4).
4. **Convergence Account.** We reverse-engineer how frontier million-token systems stack positional, KV, and serving techniques, and reframe the open problems around attention, evaluation, and data (§5).

The remainder proceeds through the three constraints in turn and closes with paired open problems, future directions, and a re-framing conclusion.

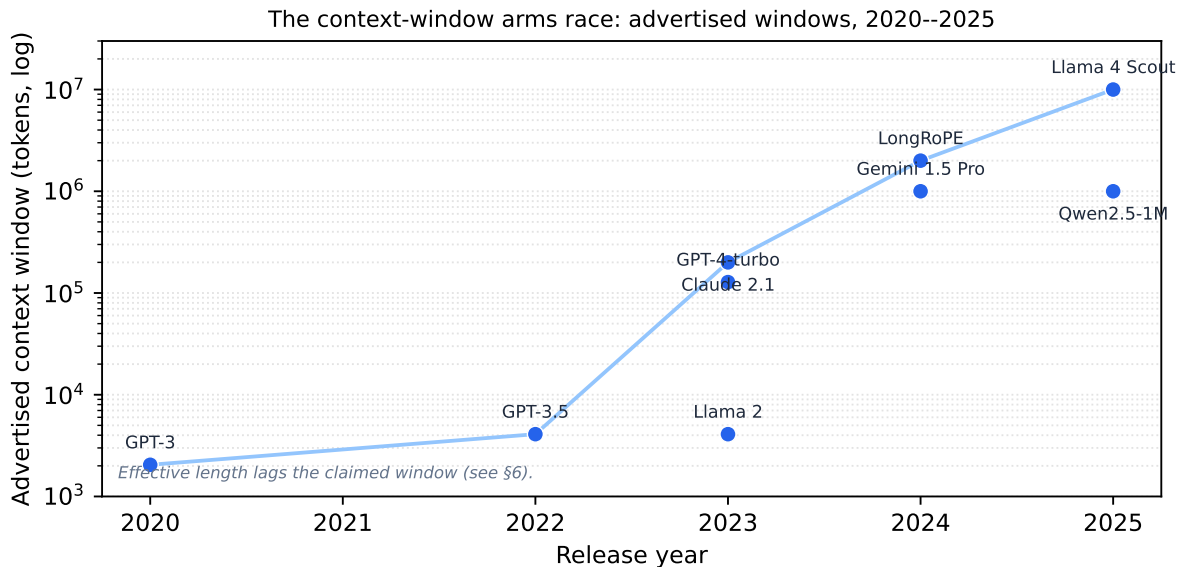


Figure 1: Advertised maximum context windows of frontier models, 2020–2025 (log scale). The claimed ceiling climbed roughly three orders of magnitude in five years; the *effective* length usable on length-controlled tasks lags far behind (§6), which is the gap this survey argues now governs the field.

2 Background: The Three Root Constraints

Long-context extension is best understood not as a catalogue of tricks but as a set of attacks on three root constraints—positional extrapolation, attention compute, and KV-cache memory—and this decomposition, not the conventional family labels, is what later lets us locate the binding constraint.

A family-based taxonomy—positional interpolation, sparse attention, KV compression, retrieval, hybrids—is the conventional organization and is arguably more neutral, because it groups methods by what they look like rather than by a causal story that could be wrong. If the three constraints were not cleanly separable, the decomposition would be a rhetorical device rather than an analytic one.

2.1 R1, R2, R3: a decomposition of the problem

(R1) Positional extrapolation. A decoder trained to length L has only observed relative positions up to L ; at inference, rotary embeddings must represent rotation angles outside the trained distribution. **(R2) Attention compute.** Dense self-attention is $\Theta(n^2d)$, so doubling the context quadruples prefill FLOPs and inflates time-to-first-token. **(R3) KV-cache memory.** Autoregressive decoding caches a key and value per token per layer; at long context this cache, not the parameters, dominates accelerator memory. Reading the literature through this lens shows R1 largely solved while R2 and R3 are where serving cost—and therefore effort—now concentrates. The three constraints differ in character: R1 is a one-time *representational* fix that, once made, stays made, whereas R2 and R3 are *recurring* per-token costs paid on every forward pass for the life of a deployment. A field optimizing total cost of ownership will keep migrating effort toward the recurring constraints long after the one-time one is settled—exactly the migration this survey

Context-window generation $\times$ feature dimensions

	Dominant lever	Positional method	KV / attention technique	Serving technique	Effective-length limiter
T1 Short-context (2k-4k...)	• none	• RoPE/absolute	• dense	• none	• training length
T2 Interpolated (8k-32k...)	• positional	• PI, NTK, YaRN	• dense	• FlashAttention	• positional...
T3 Long (128k, 2023-2024)	• positional + data	• LongRoPE	• dense / early eviction	• PagedAttention	• attention allocation
T4 Million-token (1M...)	• KV + systems	• RoPE-scaling...	• MLA, SnapKV, KIVI	• chunked prefill...	• KV memory + data
T5 Multi-million...	• sparse attention + KV	• RoPE-scaling	• NSA, MoBA, hybrids	• ring / sequence...	• long-data scarcity
<i>Key Insight: The dominant lever migrates from positional encoding (T2) to KV-cache and inference systems (T4-T5), while the positional method becomes commoditized.</i>					

Figure 2: Comprehensive taxonomy of Long-Context Extension Methods for Pretrained Language Models across 5 feature dimensions and 5 tiers (169 papers surveyed).

documents.

2.2 RoPE and why direct extrapolation breaks

Rotary position embeddings inject position by rotating query and key subvectors across a spectrum of frequencies, so attention depends on relative offset. The failure beyond the trained length is mechanical: the highest-frequency dimensions complete many rotations within L , and querying larger offsets drives their relative angles into ranges the model never fit, collapsing attention scores. Interpolation rescales offsets back into the trained range, trading angular resolution for range; NTK-aware and YaRN refine this per frequency band rather than uniformly. The point for our thesis is that this is a first-order solved problem: once offsets are representable, the residual long-context failures are not angular.

2.3 Claimed window versus effective context length

We distinguish the *claimed* window—the maximum input the architecture accepts—from the *effective* context length, the longest input at which the model still uses distant tokens reliably on a controlled task. The two diverge sharply: synthetic probes saturate while compositional tasks degrade earlier, and accuracy is U-shaped in the position of relevant information. Throughout, we treat the claimed number as a marketing figure until an effective number on a length-controlled benchmark confirms it, and we mark non-comparable cells in our matrix rather than importing self-reported bests. This distinction is not pedantry: it changes which method looks best. A technique that pushes the claimed window from 128k to 1M while leaving the effective length near 64k has done nothing for the reader who actually needs 200k-token reasoning, yet it will headline a stronger

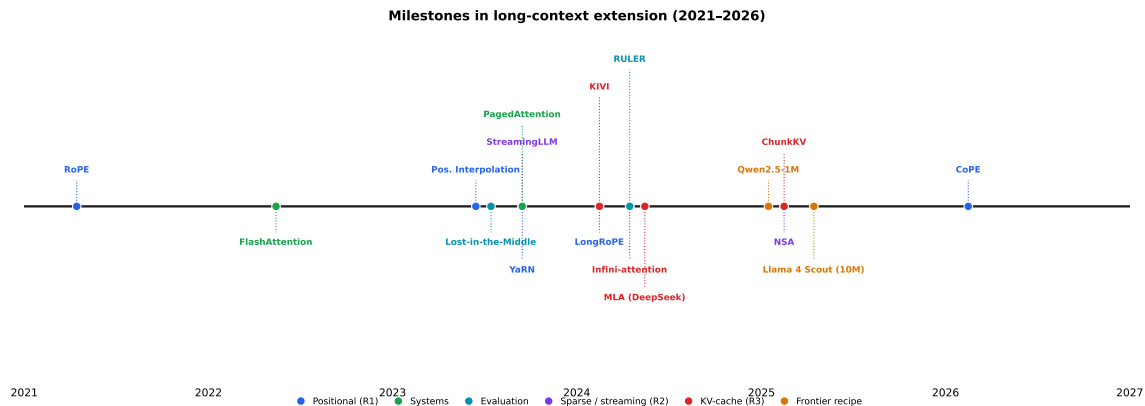


Figure 3: Milestone timeline of representative long-context methods (2021–2026), coloured by family. Reading left to right, the centre of gravity shifts from positional interpolation (blue) toward KV-cache (red), sparse attention (purple), and systems (green) work, with frontier recipes (gold) stacking all of them—the convergence this survey argues.

number than a KV scheme that quietly makes a real 128k usable and cheap to serve. Ranking methods by claimed window rewards exactly the wrong work, which is why every comparison in this survey is anchored to effective length wherever a controlled measurement exists.

2.4 What each family concedes

Reading the families against the three constraints exposes a consistent pattern of trade-offs that the rest of the survey will quantify. Positional interpolation buys range by spending angular resolution, and concedes nothing on compute or memory—which is also why it cannot help with them. Sparse and sliding-window attention buy $O(n)$ or near-linear compute by conceding access to some token pairs, risking missed long-range dependencies. KV eviction and quantization buy memory by conceding fidelity, with a quality cliff that arrives earlier on aggregation than on retrieval. Compressive-memory and retrieval-in-context methods buy unbounded length by conceding random-access precision into a lossy summary. Inference-systems engineering buys throughput and cost by conceding nothing in the model—but for the same reason cannot raise effective length on its own. Laid side by side, the trade-offs say the quiet part plainly: the only family whose concession is “nothing on R2/R3” is also the only one that cannot move the constraints that now bind.

The three constraints are not perfectly orthogonal: hybrid designs attack R2 and R3 at once, compressive memory blurs attention and cache, and a positional choice can change how attention allocates mass. The partition is analytic, not a claim that any method touches exactly one constraint.

Reading each method by which constraint it attacks, and what it concedes, is precisely what allows the following sections to show that the binding constraint has moved off positional encoding.

3 Positional-Encoding Interpolation (R1)

The RoPE-interpolation lineage—position interpolation, NTK-aware scaling, YaRN, and LongRoPE, together with training-free variants such as Self-Extend and Dual Chunk Attention—solved the positional-extrapolation failure (R1) to first order, and its marginal returns have flattened into a near-commoditized layer.

The strongest case against this claim is that positional work still moves the frontier visibly: LongRoPE reaches a *claimed* two million tokens [Ding et al., 2024], an order of magnitude beyond what KV tricks alone unlock, and new positional variants continue to appear at a steady cadence [Zhu et al., 2024]. If each generation of positional method raised effective length and not merely the claimed window, then investing in better interpolation would remain the highest-leverage move and our thesis would be wrong.

3.1 Position interpolation and NTK-aware scaling

The foundational move is to rescale position indices so that offsets beyond the trained length map back into the trained angular range, trading angular resolution for range. NTK-aware scaling improves on uniform interpolation by scaling low and high frequencies differently, so that high-frequency dimensions—which encode local order—are perturbed least. These methods are cheap and often require only light finetuning.

3.2 YaRN: per-dimension frequency treatment

Peng et al. [2023] make the per-frequency intuition precise. The key observation is that RoPE dimensions span a wide range of wavelengths: high-frequency dimensions complete a full rotation within a handful of tokens and encode fine local order, whereas low-frequency dimensions turn slowly and encode coarse global position. Uniform interpolation punishes both equally, blurring the local order the model most relies on. YaRN instead leaves short-wavelength dimensions almost untouched, interpolates only the long-wavelength dimensions that actually exceed the trained range, and adds a small attention-temperature correction to compensate for the change in score distribution. It reports efficient extension with roughly 10× fewer tokens and training steps than earlier recipes [Peng et al., 2023], and has become the de-facto open standard precisely because it makes R1 cheap; community explainers now treat RoPE-scaling as a settled, recipe-level step rather than a research frontier [EleutherAI, 2023]. The training-time variant PoSE drives the point home: it extends LLaMA to a 128k window while training on only a 2k context [Zhu et al., 2023]. The lesson generalizes: the residual error left by a good interpolation is concentrated in the slow dimensions and is small, which is why successive positional variants buy less and less.

3.3 LongRoPE and the push to two million

Ding et al. [2024] search the per-dimension rescaling factors (rather than hand-setting them) and combine a short and long context schedule to reach a claimed 2M-token window. The achievement is real, but the claimed window outruns the length at which the model is demonstrably reliable—an instance of the claimed-vs-effective gap we quantify in §6.

3.4 Training-free extension

Jin et al. [2024] (Self-Extend) reuse the model’s own attention with a grouped, bi-level position map to extend context with *no* gradient updates, and An et al. [2024] (Dual Chunk Attention) recompose attention over chunks to scale length training-free. That credible extension is now obtainable for free is the clearest signal that R1 has been commoditized.

3.5 Alternatives and refinements

Not every positional route rescales RoPE. Additive-bias schemes such as ALiBi extrapolate by penalizing distant pairs with a fixed linear bias, trading some in-window expressivity for length-robustness; they remain a useful baseline but have not displaced interpolated RoPE in frontier open models. Collinear-constrained variants such as CoCA [Zhu et al., 2024] attack a subtler failure—the anomalous behaviour of the initial-token rotation—by constraining the geometry between query and key, illustrating that the residual positional issues are now narrow and special-cased rather than fundamental. The very specificity of these fixes is itself evidence that the broad positional problem is closed. The 2026 vintage makes the point sharply: CoPE simply *clips* the rotary frequencies to extend context as a training-free “free lunch” [Li et al., 2026]—a one-line geometric tweak, not a new research programme.

3.6 Evidence of flattening returns

Across these methods the *mechanism* of improvement is the same—map offsets into a representable range—and successive variants trade diminishing accuracy for more range. The pattern is telling: PI bought the first large jump, NTK-aware and YaRN recovered most of its lost resolution, and LongRoPE’s search extracts the last increments of range at rapidly rising engineering cost while the *effective* length it unlocks lags ever further behind. A field whose latest advances are search procedures over a solved parameterization, and whose gains are increasingly claimed rather than effective, is a field in diminishing returns. The residual long-context errors these methods leave are not angular but allocational, which the next section takes up.

There is also a principled ceiling here. A positional scheme specifies *how a position is represented*, not *how attention distributes mass over represented positions*; once offsets are in-distribution, any remaining failure to use the middle of a long context, or to integrate evidence spread across it, is a property of the learned attention map and the data it was trained on, neither of which a re-parameterized rotation can touch. This is why we expect—and the evaluation section shows—that even a perfect positional method would leave the claimed-versus-effective gap largely intact. Positional work can make a length *representable*; it cannot make it *usable*. Table 1 summarises the family: every entry leaves the decode path unchanged, and the methods with the largest claimed reach are precisely those whose effective length most clearly lags.

Table 1: Comparison of Positional-Encoding Interpolation (R1) across surveyed works.

Reference	max claimed	base demo	adaptation	key limitation	mechanism
Peng et al. [2023]	128k	Llama 2	light finetune	mid-context decay	per-dim NTK RoPE scaling
Ding et al. [2024]	2M	Llama 2/Mistral	search + finetune	effective << claimed	searched per-dim rescaling
Jin et al. [2024]	~4-16x base	Llama/Mistral	training-free	extreme-length decay	grouped bi-level positions
Zhu et al. [2023]	128k	Llama	finetune (short seq)	needs train-time skips	positional skip-wise train
An et al. [2024]	~8x base	Llama 2	training-free	cross-chunk linking	dual chunk attention
Zhu et al. [2024]	extrapolative	Llama	finetune	narrow fix (init token)	collinear-constrained attn
Han et al. [2023]	zero-shot long	Llama/GPT-J	training-free	no true long-range recall	Lambda mask + distance cap

A sound positional scheme remains strictly necessary: without frequency-aware interpolation nothing downstream functions at all, and training-free variants still trail trained LongRoPE-class recipes at the most extreme lengths. Our claim is not that positional encoding is unimportant but that it has stopped being the *binding* constraint.

Once interpolation is good enough to obtain for free, differentiation must come from how attention is allocated and how the KV cache is managed—the families we turn to next.

4 Sparse Attention and KV-Cache Management (R2, R3)

Sparse and streaming attention together with KV-cache compression, eviction, and quantization carry the field’s measurable wins, and the attention-sink phenomenon shows that long-context breakdown is a softmax mass-allocation pathology rather than a positional one.

A skeptic can fairly call KV-cache compression a band-aid: it reduces the memory cost of serving a window the model already nominally supports, but it does not teach the model to *understand* more context, and aggressive eviction or quantization visibly degrades on tasks that aggregate across the whole input. On this view, KV work is engineering amortization, not progress on long context itself.

4.1 Sliding-window and learned sparse attention

Fixed-window and block-sparse attention attack R2 by computing only a subset of pairs, on the premise that most long-range attention mass is itself sparse. The design axis is *how* the retained pairs are chosen. Static patterns (local windows plus a few global tokens) are cheap but blind to content; learned and dynamic patterns spend a little computation to find the blocks that matter. Recent learned-sparse designs push this choice into pretraining: Yuan et al. [2025] (NSA) co-design a natively trainable sparse pattern with hardware-aligned kernels so the sparsity is realized as actual speedup rather than theoretical FLOP reduction, and Lu et al. [2025] (MoBA) route queries to a sparse set of key blocks, reporting strong long-context quality at a fraction of dense cost. A complementary line keeps attention dense at training time but *sparsifies the prefill* of very long prompts dynamically, exploiting the empirical concentration of attention to skip most of the quadratic work [Jiang et al., 2024, Lou et al., 2024]. The shared lesson is that the long-context regime is sparse enough that computing all n^2 pairs is wasteful—an R2 statement—and the open question is only how cheaply the important pairs can be located. Surveys of sparse attention now catalogue the growing space of answers [Bai, 2024].

4.2 Attention sinks: an allocation pathology, not a positional one

Xiao et al. [2023] observe that autoregressive models dump large attention mass onto the first few tokens—“attention sinks”—and that simply *keeping* those tokens while sliding the window restores stable generation over effectively unbounded streams—reporting up to a $22.2\times$ speedup over sliding-window recomputation [Xiao et al., 2023]—with no retraining and no positional change. The mechanism is instructive: softmax over keys must sum to one, so when no key is genuinely relevant the model must still place its attention mass somewhere, and it learns to park that surplus on a few always-present early tokens that act as a no-op drain. Slide the window so those drain tokens fall out of scope and the softmax has nowhere to dump the surplus, the score distribution destabilizes, and perplexity explodes—even though every position index is still in range. This is decisive for our thesis: the failure that sinks fix is about where softmax mass goes, not about position indices, which is why an inference-time allocation trick—not a better RoPE—repairs it. The same allocation view explains the lost-in-the-middle curve of §6: mass concentrates at the edges because that is where the model has learned to look, not because middle positions are mis-encoded.

4.3 KV eviction

A family of methods keeps only the keys/values that matter. Zhang et al. [2023] (H2O) evict all but “heavy-hitter” tokens using accumulated attention, and Liu et al. [2023b] exploit the persistence of importance across decode steps. Working at prefill instead, Li et al. [2024b] select salient KV from observed query patterns, reporting a $3.6\times$ generation speedup and $8.2\times$ better memory efficiency, while Yang et al. [2024] taper the KV budget across layers. Each recovers most quality at a large memory cut, training-free. The 2025 frontier refines *what unit* is evicted: ChunkKV compresses at the granularity of semantic chunks rather than isolated tokens, preserving inter-token structure at high compression [Liu et al., 2025], and TokenSelect makes the selection dynamic and query-dependent, coupling eviction to length extrapolation [Wu et al., 2025].

4.4 KV quantization and architectural compression

Orthogonally, Liu et al. [2023c] quantize the KV cache to 2 bits with asymmetric per-channel handling, and Gholami et al. [2024] push toward sub-4-bit KV explicitly targeting ten-million-token inference. DeepSeek’s multi-head latent attention [DeepSeek-AI et al., 2024a] bakes the compression into the architecture, projecting KV into a small latent space so the cache shrinks by construction.

4.5 Compressive memory and retrieval-in-context

The KV cache is the dominant decode-time memory cost at long context, the shared target of every method in this family [Raschka, 2025b]. A third route stores the past in bounded memory or retrieves it on demand: Munkhdalai et al. [2024] (Infini-attention) fold a compressive memory into standard attention for unbounded inputs at bounded cost, Zhang et al. [2024] (Activation Beacon) compress activations into summary tokens, and Mohtashami and Jaggi [2023] attach landmark tokens for random access. The latest systems take the retrieval framing literally: RetroInfer treats the KV cache as a vector store and fetches only the relevant entries at decode, recasting long-context inference as a storage-and-retrieval problem [Chen et al., 2026]. Table 2 compares this family along its own axes—KV mechanism, compression form, and whether it is training-free—and makes the family’s defining trait explicit: the large majority operate *training-free* at inference, in sharp contrast to the positional family’s reliance on finetuning. Table 3 then places all of these against the positional family along our nine dimensions.

Read down the matrix’s root-cause column and a pattern emerges that the prose alone obscures. The positional family attacks R1 and leaves the decode profile untouched—its rows all read “dense; no KV change”—so whatever it does for reach, it does nothing for the cost of serving that reach. Every row that actually moves the decode-and-KV column belongs to the sparse-attention or KV family, and those are also the rows whose adaptation cost is most often “training-free”. The matrix thus encodes the survey’s argument structurally: the methods that change what it costs to use a long context, and that can be applied without retraining, are precisely the non-positional ones. The “reach: claimed / effective” column carries the complementary warning—wherever the two figures diverge sharply, as for the 2M-token positional entry, the row is flagged by its primary failure mode of “effective $\ll$ claimed” rather than by any decode-time virtue.

Table 2: Comparison of Sparse Attention and KV-Cache Management (R2, R3) across surveyed works.

Reference	training free	compression	key limitation	kv mechanism
Xiao et al. [2023]	yes	$O(1)$ window KV	no long-range recall	streaming + attn sinks
Zhang et al. [2023]	yes	$\sim 5\text{-}10\times$ KV cut	aggregation-task drop	heavy-hitter eviction
Li et al. [2024b]	yes	large KV cut	needs query locality	prefill salient selection
Yang et al. [2024]	yes	layer-wise cut	deep-layer info loss	layer-tapered KV budget
Liu et al. [2023b]	yes	large KV cut	needs importance stability	persistence eviction
Liu et al. [2023c]	yes	$\sim 2.6\times$ (2-bit)	outlier channels	2-bit asym KV quant
Gholami et al. [2024]	calibration	$< 4\text{-bit}$ KV	calibration drift	sub-4-bit KV quant
Munkhdalai et al. [2024]	no (cont. pretrain)	bounded memory	fixed-capacity forgetting	compressive memory
Zhang et al. [2024]	no (finetune)	summary tokens	lossy summarization	activation-beacon summaries
Mohtashami and Jaggi [2023]	no (finetune)	block index	block granularity	landmark random access
Tworowski et al. [2023]	no (finetune)	retrieved KV	distractor sensitivity	contrastive retrieval
Yuan et al. [2025]	no (pretrain)	sparse KV	rare-token misses	native sparse attention
Lu et al. [2025]	no (finetune)	block-sparse KV	block-granularity loss	block-attention routing
DeepSeek-AI et al. [2024a]	no (pretrain)	$\sim \text{KV}/8$ latent	from-scratch design	latent attention (MLA)

Table 3: Cross-cutting comparison of sixteen influential long-context methods. Root cause: R1 positional extrapolation, R2 attention compute $O(n^2)$, R3 KV-cache memory $O(n)$. “Reach” separates the *claimed* window from the length at which the method is *demonstrably* helpful. Effective figures are approximate and benchmark-dependent; see §6.

Method	Family / root cause	Reach: claimed / effective	Adaptation	Decode & KV profile	Primary failure mode
PI / NTK-aware	positional interp. (R1)	8–32k / moderate	finetune or free	dense; no KV change	resolution crowding
YaRN [Peng et al., 2023]	positional interp. (R1)	128k / 64–128k	finetune ($\sim 0.1\%$ tokens)	dense; no KV change	mid-context decay
LongRoPE [Ding et al., 2024]	positional interp. (R1)	2M / $\ll 2$ M	search + finetune	dense; no KV change	effective $\ll$ claimed
Self-Extend [Jin et al., 2024]	positional, training-free (R1)	4–16 $\times$ base / moderate	training-free	dense; grouped attn	extreme-length decay
Dual Chunk [An et al., 2024]	positional, training-free (R1)	8 $\times$ base / good	training-free	dense; chunked	cross-chunk linking
StreamingLLM [Xiao et al., 2023]	streaming attn. (R2,R3)	∞ stream / window-bound	training-free	$O(1)$ KV via sinks	no true long-range recall
NSA [Yuan et al., 2025]	native sparse attn. (R2)	64k+ / strong	pretrain/finetune	sparse; reduced KV	rare-token misses
MoBA [Lu et al., 2025]	block sparse attn. (R2)	1M / strong	finetune	block-sparse KV	block-granularity loss
H2O [Zhang et al., 2023]	KV eviction (R3)	inherits base / good	training-free	$\sim 5\text{-}10\times$ KV cut	aggregation-task drop
SnapKV [Li et al., 2024b]	KV eviction (R3)	inherits base / strong	training-free	large KV cut at prefill	needs query locality
PyramidInfer [Yang et al., 2024]	layer-wise KV (R3)	inherits base / good	training-free	layer-tapered KV	deep-layer info loss
KIVI [Liu et al., 2023c]	KV quantization (R3)	inherits base / strong	training-free	2-bit KV ($\sim 2.6\times$)	outlier channels
KVQuant [Gholami et al., 2024]	KV quantization (R3)	10M target / strong	calibration	sub-4-bit KV	calibration drift
MLA [DeepSeek-AI et al., 2024a]	architectural KV (R3)	128k / strong	pretrain	latent KV ($\sim \text{KV}/8$)	needs from-scratch design
Infini-attn. [Munkhdalai et al., 2024]	compressive memory (R2,R3)	∞ / task-dependent	continued pretrain	bounded memory	fixed-capacity forgetting
FlashAttention [Dao et al., 2022]	systems / kernel (R2)	enabler / n.a.	training-free	IO-aware; exact	no semantic change

4.6 A unifying view of KV management

The four KV strategies above—eviction, quantization, architectural compression, and compressive memory/retrieval—look disparate but reduce to a single question: *which information in the cache can be discarded or coarsened without changing the next-token distribution?* Eviction answers “the tokens that attention rarely selects”; quantization answers “the low-order bits of every entry”; architectural latent attention answers “everything outside a learned low-rank subspace”; and compressive memory answers “the fine detail of the distant past, kept only in summary.” Seen this way, the family is one idea explored along different axes (which tokens, how many bits, what rank, what summary), and its progress is measured by how aggressively the cache can be cut before the distribution shifts—a quantity that, unlike claimed window length, has steadily improved year over year. That steady, measurable improvement is the empirical heart of our thesis.

The skeptic’s point partly stands: aggressive compression does have quality cliffs at extreme ratios and on aggregation or multi-hop tasks, and sparse patterns can miss genuine long-range dependencies. These are real costs, recorded in the matrix’s failure-mode column.

That near-lossless, high-ratio KV compression is achievable *at all* is precisely the measurable gain positional methods no longer deliver, which relocates the frontier to attention allocation and the systems that serve it.

5 Inference Systems and Production Recipes

Inference-systems engineering is what actually makes long context shippable and cheap, and frontier 1M-token recipes stack these techniques rather than introduce a novel positional trick.

One can argue systems work is mere plumbing: FlashAttention and PagedAttention change *how fast* and *how cheaply* a window runs, not *whether* the model understands it, so the real capability was created upstream by architecture and positional pretraining and systems merely amortize it.

5.1 IO-aware kernels

Dao et al. [2022] make exact attention IO-aware. The bottleneck in a naive implementation is not arithmetic but memory traffic: materializing the full $n \times n$ score matrix in high-bandwidth memory costs $O(n^2)$ reads and writes that dominate wall-clock time. FlashAttention tiles the computation so that blocks of queries and keys are streamed through fast on-chip SRAM and the softmax is computed in a numerically stable online pass, never writing the large matrix to slow memory. The result is exact attention—no approximation—at a fraction of the memory traffic, which is what turned long-context prefill from impractical to routine. Because the change is purely in the schedule, the technique composes with every method in the matrix: it changes the cost of attention without changing its semantics, and it is the substrate the entire serving stack assumes.

5.2 Paged and reused KV

Kwon et al. [2023] (PagedAttention/vLLM) manage the KV cache like virtual memory: rather than reserving one contiguous slab per sequence, the cache is split into fixed-size blocks mapped through an indirection table, which eliminates the internal fragmentation that previously wasted 60–80% of cache memory and—because blocks can be shared—lets many requests reuse the key-values of a common system prompt. Prefix caching generalizes this to persist and reuse the KV of any repeated prefix across calls, turning a recomputation into a lookup, and Ye et al. [2024] make that sharing explicit with a prefix-aware KV layout. Chunked prefill interleaves long-prompt prefill with ongoing decode so a single giant prompt does not stall the whole serving batch. Jiang et al. [2024] attack the same prefill cost from the algorithmic side: a 1M-token prompt that otherwise takes about 30 minutes to pre-fill on an 8B model is accelerated by exploiting dynamic attention sparsity to skip most of the quadratic work. None of these changes the model’s weights or its positional scheme; they change only the cost and scheduling of serving a window the model already supports—which is exactly why they are where the marginal engineering now goes.

5.3 Parallelism for million-token prefill

Liu et al. [2023a] (Ring Attention) distribute a single long sequence across devices with block-wise computation and overlapped communication, making near-million-token contexts tractable; sequence/Ulysses parallelism generalizes this. Zhao et al. [2025] document the systems engineering—memory, communication, load balance—behind a frontier deployment.

5.4 Production recipes: what frontier 1M models actually stack

Reverse-engineering shipped systems is revealing. Qwen’s recipe [Bai et al., 2023] and DeepSeek-V3 [DeepSeek-AI et al., 2024b] combine grouped/latent attention, RoPE scaling, two-stage long-context training, and a heavy serving stack. Qwen2.5-1M’s own deployment write-up makes the division of labour explicit: the 1M window is reached by RoPE-scaling plus length training, but it is only *usable* through a chunked-prefill, sparse-attention serving stack, and an agentic chunk-and-retrieve fallback stretches effective context further still [Qwen Team, 2025, 2024]. DeepSeek’s contribution at long context is MLA [DeepSeek-AI et al., 2024a]—a *KV* idea, not a positional one, shipped in a 236B-parameter (21B-active) model with a 128K window—and its successor doubles down on sparse attention rather than new position encodings [Raschka, 2025a]. Hybrid attention models such as Jamba [Lieber et al., 2024] interleave attention with state-space layers to bound the KV footprint. Across these systems the positional method is interchangeable and commoditized; the differentiation is in KV and serving.

5.5 The economics that drive the shift

Why the marginal effort flows to systems is, at bottom, an economic argument. At long context the decode-time cost is dominated not by matrix multiplies but by reading the KV cache from memory: every generated token must stream the entire cache, so per-token latency and the achievable batch size are set by cache size, not by the positional scheme. A method that halves the KV footprint therefore roughly doubles serving throughput and halves the cost per token, whereas a better interpolation leaves the serving bill untouched. This is why prefix caching, paging, quantization, and latent-KV attention—each a direct attack on cache traffic—are the levers a provider actually pulls, and why the comparison matrix’s “decode & KV profile” column, not its “reach” column, predicts which methods ship. The positional layer is free; the cache is the meter that runs.

The frontier of this economic logic is to stop holding the whole cache in fast memory at all. Disaggregated and offloaded serving moves cold KV to host memory, NVMe, or a separate store and pages it back on demand, while retrieval-style engines treat the cache as a vector index and fetch only the entries a query needs [Chen et al., 2026]. These designs reframe long-context inference as a storage-and-retrieval problem rather than an attention problem—a move that is only sensible once one accepts that the binding constraint is cache traffic, not positional representation. That the newest systems work organizes itself around the cache, not the rotation, is perhaps the clearest signal of where the field believes the remaining gains live.

Table 4: Comparison of Inference Systems and Production Recipes across surveyed works.

Reference	layer	model change	optimizes	key idea
Dao et al. [2022]	kernel	none (exact)	attention memory traffic	IO-aware tiled softmax
Kwon et al. [2023]	serving	none	KV memory + sharing	paged KV, prefix reuse
Liu et al. [2023a]	parallelism	none	million-token prefill	blockwise ring attention
Jiang et al. [2024]	serving	none	long-prompt prefill	dynamic prefill sparsity
Ye et al. [2024]	serving	none	shared prefix KV	prefix-aware KV layout
DeepSeek-AI et al. [2024b]	recipe	MLA + sparse	train + serve at scale	stacked positional+KV+sys
Bai et al. [2023]	recipe	GQA + RoPE-scale	long-context deploy	2-stage long training
Lieber et al. [2024]	architecture	attn+SSM hybrid	bounded KV footprint	interleave attn/state-space
Zhao et al. [2025]	recipe	MLA	frontier deployment	memory/comm/load balance

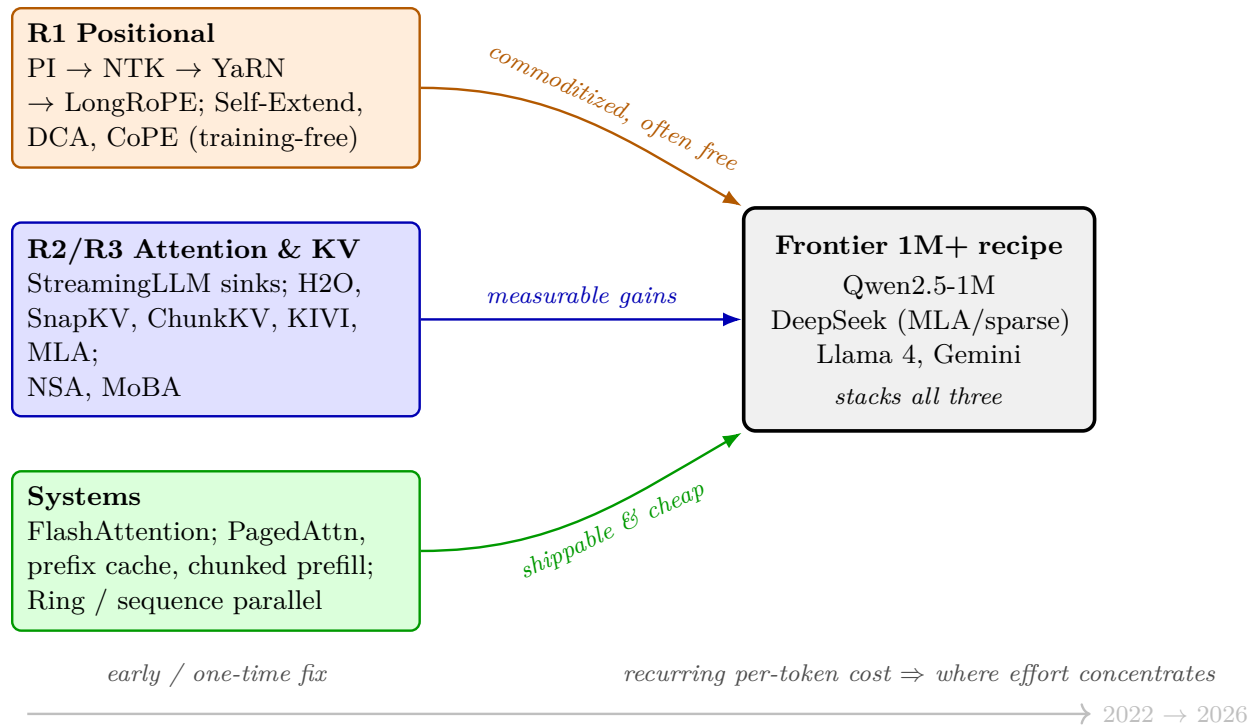


Figure 4: Convergence of the long-context families. The positional family (R1) is a one-time, increasingly training-free fix; the attention/KV (R2/R3) and systems families carry the recurring, measurable gains. Frontier million-token systems stack all three, with positional encoding as commoditized table stakes rather than the differentiator—the trajectory this survey’s thesis predicts.

5.6 Convergence

Figure 4 draws the argument directly: the three families flow into a single stacked production recipe, with the positional layer entering as commoditized table stakes while the attention/KV and systems layers carry the gains that make a million-token window both correct and affordable. Table 4 underlines that not one systems entry changes the model’s positional scheme—they change only what it costs to serve. Read alongside it, Figure 2 reads as a trajectory: the dominant lever migrates from positional encoding at 8–32k to KV-cache and inference systems at 1–10M, and the milestone timeline of Figure 3 shows the same shift in which family’s landmark works dominate each period—exactly the convergence our thesis predicts.

5.7 When to use what: a decision guide

The three-constraint frame yields an actionable recipe selector keyed to the regime a practitioner is in. **Up to 32k, compute-bound prefill:** a sound RoPE scaling (YaRN) plus FlashAttention is sufficient; KV memory is not yet the bottleneck, so spend nothing on eviction. **32k–128k, memory-pressured serving:** keep the positional recipe fixed and add training-free KV management—SnapKV or H2O for eviction, KIVI for quantization—plus PagedAttention and prefix caching, which is where cost is now decided. **128k–1M, throughput-critical production:** adopt architectural KV compression (MLA) or a learned/sparse attention pattern (NSA, MoBA), pair it with chunked prefill and sequence parallelism, and treat the positional layer as a solved

commodity. **Beyond 1M, or when effective length matters more than the headline:** the binding constraint is data and allocation, not encoding, so invest in long-dependency training data, attention-allocation fixes, and retrieval/offload serving rather than another positional variant. The through-line is that the right move is read off the *binding constraint at that length*—which, past 32k, is almost never positional.

Systems engineering does not by itself raise effective context: it lowers the cost of serving whatever window the model already supports, and a poorly trained long model stays poor however efficiently it is served.

Because the marginal dollar and the marginal token now flow to KV systems, that is where the field’s effort and differentiation have concentrated—a shift the evaluation evidence in the next section explains.

6 Evaluation: The Claimed-vs-Effective Gap

Length-controlled benchmarks reveal a large claimed-vs-effective gap that does not track which positional variant a model uses, implicating attention allocation and long-data scarcity—not position encoding—as the cap on usable context.

If benchmarks merely lag real capability, the gap is an artifact: needle- in-a-haystack retrieval is near-saturated for frontier systems, so one could argue the windows are genuine and only the probes are weak, in which case better positional scaling would still be the right investment.

6.1 NIAH and its saturation

Single-needle retrieval is now easy for strong long-context models, which is precisely why it is a poor proxy: a model can copy a planted sentence without integrating the surrounding context, so saturation on NIAH says little about effective context on real tasks.

6.2 Controlling length and task

Hsieh et al. [2024] (RULER) vary sequence length and task complexity systematically. Its task families are diagnostic by design: multi-key and multi-value needle retrieval probe whether the model can disambiguate among many planted facts; *variable tracking* requires following a chain of assignments across the context, so a model cannot succeed by local copying; and *aggregation* tasks (e.g. counting or common-word extraction) demand integrating evidence spread over the whole input. The empirical finding is that models advertising large windows sustain quality on single-needle retrieval far past the length at which their variable-tracking and aggregation scores collapse—so the “effective length” depends on which capability one measures, and the honest summary is the length at which the *hardest* task still clears threshold. Bilingual, multitask suites such as LongBench [Bai et al., 2024] and multilingual extensions [Kim et al., 2025] corroborate the pattern on realistic inputs, and show the gap widens in lower-resource languages where long-dependency training data is scarcer—a hint that data, not geometry, is binding.

6.3 Operationalizing effective length

To make “effective context length” more than a slogan we adopt an explicit definition: the largest input length L^* at which a model’s accuracy on a fixed task stays above a threshold (commonly the mid-eighties percent on RULER-style tasks) of its short-context accuracy. This turns a vague capability into a single comparable number and exposes two facts the advertised window hides.

First, L^* is *task-relative*: a model may clear threshold at 128k on single-needle retrieval yet fall below it at 16k on variable tracking, so a method’s honest reach is the L^* of its weakest in-scope task, not its best. Second, L^* moves with KV handling and data far more than with the positional recipe—the same base model, re-served with a different KV budget or trained on more long-dependency data, posts a different L^* under an unchanged RoPE scaling. Reporting L^* rather than the claimed window would, on its own, redirect the field toward the levers this survey argues are binding.

The two curves come apart visibly. Figure 1 plots the claimed ceiling climbing three orders of magnitude in five years; the L^* curve, were we to overlay it, would rise far more slowly and flatten where the claimed line keeps accelerating. The gap is not noise to be averaged away but the central datum: a model advertised at 1M tokens whose L^* on multi-hop tracing sits near 64k has a $15\times$ discrepancy, and that discrepancy is set by how its attention allocates and how much long-dependency data it saw, not by whether it used YaRN or LongRoPE. A reader who internalises only one figure from this survey should take that one: the headline number and the usable number are diverging, and the divergence is the field’s real frontier.

6.4 Lost in the middle and effective length

Liu et al. [2024] document a U-shaped accuracy curve: models use information at the start and end of the context well and the middle poorly. Two consequences matter for our thesis. First, this is an *allocation* signature, not a positional-index error. Second, and non-obviously, effective context length is largely independent of which RoPE-interpolation variant produced the window—two models sharing a positional recipe can differ several-fold in effective length depending on their KV handling and long-data supply, which points the causal arrow at attention and data rather than at position encoding.

6.5 Contamination and benchmark validity

Because long-context test material is often drawn from public corpora, contamination can inflate apparent skill, and saturated probes mask degradation; this is why length-controlled, compositional, dynamically generated benchmarks are displacing static NIAH as the field’s measuring instrument. ∞ Bench pushes average lengths past 100k to probe the regime where retrieval, computation, and dialogue must compose over book-length inputs, and its scores fall well below short-context base-lines even for models that ace NIAH—confirming that the easy probe and the hard regime measure different things. Table 5 summarises what each instrument measures and the failure it exposes; tellingly, the column that varies across rows is *which* failure is surfaced, not *whether* a gap exists.

Table 5: Comparison of Evaluation: The Claimed-vs-Effective Gap across surveyed works.

Reference	exposes	measures	key finding
Hsieh et al. [2024]	multi-hop/aggregation drop	length and task complexity	effective far below claimed
Liu et al. [2024]	lost-in-the-middle	position of evidence	U-shaped accuracy
Bai et al. [2024]	realistic-task gap	bilingual real tasks	degrades before claimed len
Kim et al. [2025]	data-scarcity effect	multilingual long context	gap widens low-resource

Synthetic probes are imperfect proxies, and some reported gaps reflect task design rather than a hard capacity ceiling; no single benchmark settles the question, and effective-length numbers are benchmark-relative.

Read as a diagnostic rather than a scoreboard, the evaluation literature points the field’s remaining work at attention pattern, data, and measurement validity—the agenda of our open problems.

7 Challenges and Open Problems

7.1 The claimed-vs-effective gap is not closing

Effective context length lags advertised windows by large factors, and the gap is insensitive to the positional recipe used. Length-controlled benchmarks [Hsieh et al., 2024, Liu et al., 2024] now expose the gap, and longer-window recipes [Ding et al., 2024] keep raising the claimed number. Raising the claimed window does not raise effective length; reporting practice still headlines the architectural capacity. Adopt effective length on compositional suites as the primary metric (FD1), so progress is measured where it is bottlenecked.

7.2 KV-compression quality cliffs

Aggressive eviction and quantization degrade sharply on aggregation and multi-hop tasks beyond a method-specific ratio. Heavy-hitter eviction [Zhang et al., 2023], salient selection [Li et al., 2024b], and low-bit quantization [Liu et al., 2023c] recover most quality at moderate compression. A single global budget ignores that different heads, layers, and queries need different amounts of cache, so the cliff arrives early on hard tasks. Task- and head-adaptive, learned KV budgets (FD2) that allocate memory where the computation needs it.

7.3 Attention-sink dependence and lost-in-the-middle

Models lean on a few sink tokens and underuse the middle of the context, a pattern pathology rather than a positional one. Retaining sink tokens stabilizes streaming [Xiao et al., 2023]; sparse routing [Lu et al., 2025] redistributes some mass. These manage the symptom; the underlying softmax mass-allocation bias remains, so the middle stays weak. Attention-allocation regularization during training and inference (FD3) to spread mass beyond sinks and the edges.

7.4 Long-data scarcity

There is little genuinely long-dependency supervision, so training cannot teach what evaluation demands. Two-stage long-context pretraining [DeepSeek-AI et al., 2024b] and synthetic needle data are the current stopgaps. Synthetic retrieval data teaches copying, not long-range reasoning, and natural long-dependency text is rare. Compositional long-dependency curricula (FD4) that manufacture the signal natural corpora lack.

7.5 Benchmark contamination and NIAH saturation

Easy retrieval probes saturate and public-corpus contamination inflates apparent long-context skill. RULER [Hsieh et al., 2024] and bilingual suites [Bai et al., 2024] add task variety. Static suites are still memorizable, and saturated probes hide degradation. Contamination-resistant, dynamically generated compositional evaluation (FD5).

7.6 Unmeasured stacking interactions

Production recipes stack many techniques but the marginal contribution of each, and their interactions, are folklore. System reports describe the full stack [DeepSeek-AI et al., 2024b, Zhao et al., 2025] without isolating components. Without ablations, the field cannot tell which layer of the stack actually buys effective length. Ablation standards for composed long-context stacks (FD6).

8 Future Directions

The open problems above pair one-to-one with the directions below; each direction names a concrete next step rather than a wish.

FD1 — Effective length as the primary metric. We expect reporting to move from advertised windows to effective context length on length-controlled, compositional suites [Hsieh et al., 2024]; the trajectory of benchmark design suggests the headline number will become “effective at N ”, not “supports N ”.

FD2 — Task-adaptive, learned KV budgets. Building on per-layer and salient-token schemes [Yang et al., 2024, Li et al., 2024b], the evidence points toward budgets allocated per head, layer, and query, learned rather than hand-set, to push the quality cliff outward. This direction is already materializing: LKV learns head-wise KV budgets and token selection end-to-end [Zhou et al., 2026], exactly replacing the global hand-set ratio our analysis identified as the source of the compression cliff.

FD3 — Attention-allocation regularization. Because sinks and the lost-in-the-middle curve are allocation signatures [Xiao et al., 2023, Liu et al., 2024], training objectives and inference interventions that explicitly regularize where softmax mass goes are a likely lever on effective length.

FD4 — Synthetic long-dependency curricula. If long-data scarcity is the true cap, manufacturing compositional long-range supervision—beyond needle-style copying—should raise effective length more than further positional scaling.

FD5 — Contamination-resistant compositional evaluation. Dynamically generated, multi-hop benchmarks resistant to memorization would let the field measure genuine progress as static probes saturate.

FD6 — Ablation standards for stacked recipes. Reporting the marginal contribution of each technique in a composed stack [DeepSeek-AI et al., 2024b] would convert recipe folklore into transferable knowledge.

9 Conclusion

The context-window arms race created an illusion of progress that this survey has tried to dispel: long-context capability is no longer bottlenecked by positional-encoding interpolation, and the durable gap between a model’s claimed window and its effective context length shows that attention allocation and long-data scarcity, not position encodings, now cap usable context. Read through the three root constraints, the field’s last three years tell a single story of a moving bottleneck.

Positional encoding became table stakes. The PI→NTK→YaRN→LongRoPE lineage, and the fact that credible extension is now obtainable with zero gradient updates, mark R1 as first-order solved and commoditized rather than as the axis of differentiation.

The wins migrated to KV and attention. Eviction, quantization, architectural latent caches, and sparse attention deliver the measurable gains today, and the attention-sink result reframes long-context failure as a softmax mass-allocation pathology—an inference-time, not a positional, problem.

Serving economics decide what ships. FlashAttention, paged and reused KV, and sequence parallelism are what make million-token windows affordable, and frontier recipes stack these with a commoditized positional layer rather than a novel one—visible convergence rather than a single breakthrough.

Evaluation is the field’s compass, not its scoreboard. The claimed-vs-effective gap, and its insensitivity to the positional recipe, are only legible under length-controlled, compositional benchmarks, which is why measurement validity is now load-bearing. The most consequential number a long-context paper can report is no longer the window it accepts but the length at which it still reasons; until reporting practice makes that switch, the headline figures will keep overstating capability and misdirecting effort toward the constraint that is already solved.

The bottleneck is data and allocation, not geometry. The deepest implication of the three-constraint reading is that the next gains are unlikely to come from rotating embeddings more cleverly. Two models that share a positional recipe can differ several-fold in effective length, which locates the causal levers in how attention distributes mass and in the scarcity of genuinely long-dependency training signal—neither of which a better interpolation scheme can supply.

A call to action. The productive next moves are to report effective length rather than advertised windows, to learn KV budgets and regularize attention allocation, and to manufacture the long-dependency data the bottleneck implies. This thesis would have to be revised if a purely positional method, with no KV or systems change, were shown to close the claimed-vs-effective gap on a contamination-resistant compositional benchmark—the one experiment that would put position encoding back at the center. Until then, the honest reading is that the next order of magnitude of *usable* context will be won in the cache and the kernel, not in the rotation of the embedding.

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